

6.2 Fundamental particles

The work here concerns the structure and binding of atoms and nuclei and the nature of fundamental particles. The practical implications of radioactivity are considered, introducing the idea of risk.

The first section (6.2.1) considers scattering experiments as a source of evidence about the structure of atoms and nucleons. Ideas from earlier in the module are used to consider particle paths in magnetic and electric fields in the context of particle accelerators. Evidence for discrete energy levels leads on to a crude model of the atom as a particle

in a box. This section gives more opportunities to discuss the development of models in physics and the international cooperation needed to fund large experiments (HSW11).

The second section (6.2.2) sees changes in nuclear binding energy per nucleon as driving different types of decay. This leads to a consideration of nuclear power generation. The biological effects of ionising radiation are also considered, giving more opportunity to consider issues of ethics, decision making and the risks and benefits of technology (HSW9, HSW10, HSW12).

6.2.1 Probing deep into matter

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i>	
(i) use of particle accelerators to generate high-energy beams of particles for scattering	HSW7, 11
(ii) evidence from scattering for a small massive nucleus within the atom	Development of the nuclear model HSW7, HSW11
(iii) evidence of discrete energy levels in atoms	For example from collisions with electrons or from line spectra, HSW12
(iv) a simple model of the atom as the quantum behaviour of electrons in a confined space	HSW7
(v) simple picture of the quark structure of protons and neutrons	
(vi) application of conservation of mass/energy, charge and lepton number in balanced nuclear equations	
(vii) relativistic calculations for particles travelling at very high speed, for example in particle accelerators or cosmic rays.	
(b) <i>Make appropriate use of:</i>	
(i) the terms: energy level, scattering, nucleus, proton, neutron, nucleon, electron, positron, quark, gluon, neutrino, hadron, lepton, antiparticle, lepton number	HSW8
<i>by sketching and interpreting:</i>	
(ii) paths of scattered particles	
(iii) electron standing waves in simple models of an atom.	

(c) *Make calculations and estimates involving:*

- (i)** motion of a charged particle in magnetic field using $F = qvB$ *M2.3*
- (ii)** kinetic and potential energy of a scattered charged particle
- (iii)** $E_{\text{rest}} = mc^2$ and relativistic factor *M2.2*
HSW11, 12
- (iv)** $E_{\text{total}} = \gamma E_{\text{rest}}$.